

Thorough Investigation

1. Project Viability and Overall Assessment

The goal of this project was to explore whether an end-to-end data engineering and analytics pipeline could provide meaningful insights into Spotify song popularity using audio features. From the start, my primary interest was not just in building a final model, but in understanding the full process from data ingestion and transformation, through exploratory analysis, modeling, and visualization, and learning what the data itself could reveal at each stage.

As a pilot project, the system performs well and successfully demonstrates technical feasibility. The Airflow-managed pipeline reliably ingests raw Spotify data, applies transformations to produce a clean analytical dataset, and supports both EDA and modeling workflows. The cleaned dataset (`clean_spotify.pkl`) proved suitable for analysis, and the downstream steps ran consistently without manual intervention. Overall, this confirmed that the pipeline design is functional, extensible, and appropriate for continued experimentation.

2. Scalability and Production Considerations

While this project was intentionally scoped as a prototype, it offers a clear path toward scaling if the project were to continue.

From a data engineering perspective, the pipeline currently operates in batch mode on a static dataset. Scaling would require incremental ingestion strategies and more efficient storage formats, such as Parquet. Dataset versioning would also be important for tracking changes over time and ensuring reproducibility.

From an orchestration standpoint, Airflow was an effective choice for managing the workflow and learning how production pipelines are structured. To scale the system, additional steps such as data quality checks, schema validation, and failure alerting would be needed. Model training, which is currently run as a standalone script, could also be fully integrated into the DAG to enable scheduled retraining.

This prototype confirms that the technical foundation is solid, while also clearly highlighting what would need to change for a production environment.

3. Innovation and Learning Value

The core innovation of this project lies less in creating a novel prediction algorithm and more in the process of building and validating a complete data pipeline. My main focus was on exploring the relationships within the data, particularly how different audio features contribute to popularity, and validating those findings through multiple layers of analysis.

By combining EDA, machine learning, and visualization, the project moves beyond a simple “train a model” exercise. Feature importance from the RandomForest model was used to reinforce patterns discovered during EDA, creating an iterative feedback loop between exploration and modeling. This emphasis on interpretability allowed me to better understand why certain features mattered, rather than treating the model as a black box.

4. Technical Limitations and Challenges

Several limitations became apparent during the project, which also served as valuable learning points.

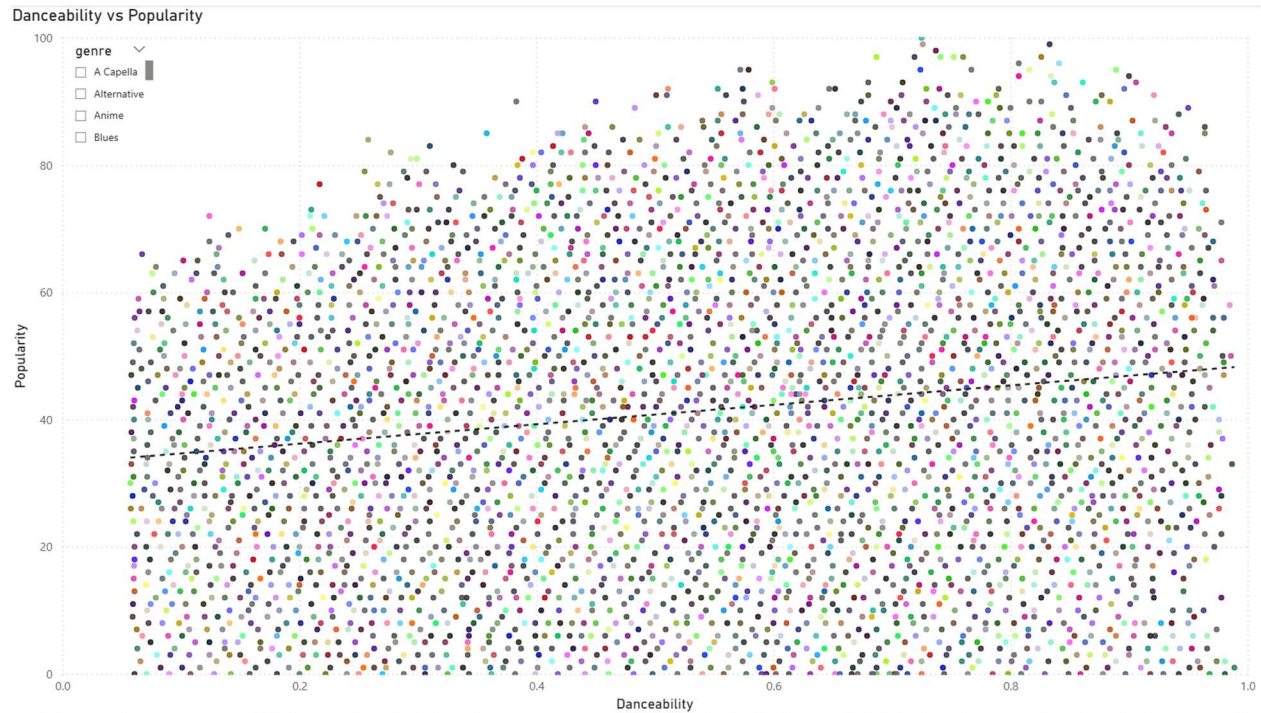
Spotify popularity is influenced by many external factors such as marketing, playlist placement, and social trends that are not captured by audio features alone. As a result, the model can explain general trends but cannot fully predict popularity. Additionally, defining a “hit” using a popularity threshold introduces subjectivity and affects class balance.

From a pipeline perspective, using pickle files was convenient for experimentation but is not ideal for long-term storage or scalability. The pipeline also assumes reasonably clean input data and lacks formal data validation steps. These limitations were acceptable for a prototype but would need to be addressed in a larger system.

5. Power BI Visualizations and Observations

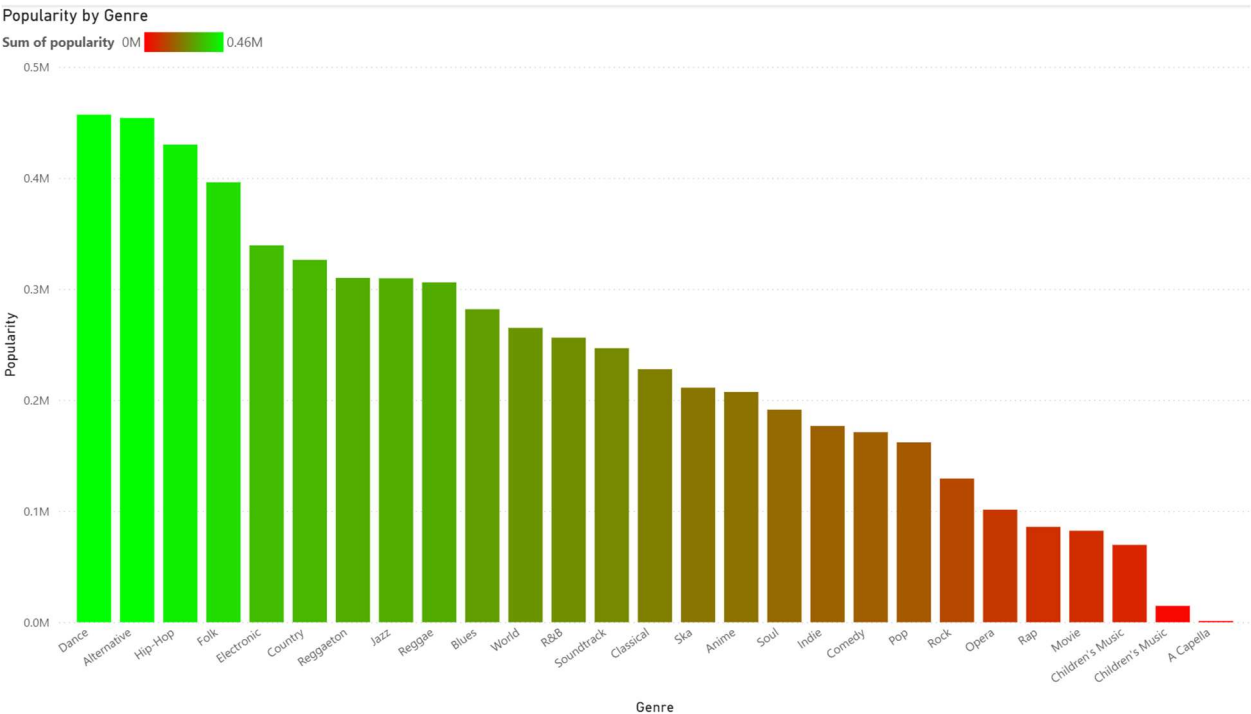
Power BI was used to explore the data visually and to communicate findings in a clear, intuitive way. The dashboard played a key role in reinforcing and contextualizing insights discovered during EDA and modeling.

Danceability vs Popularity



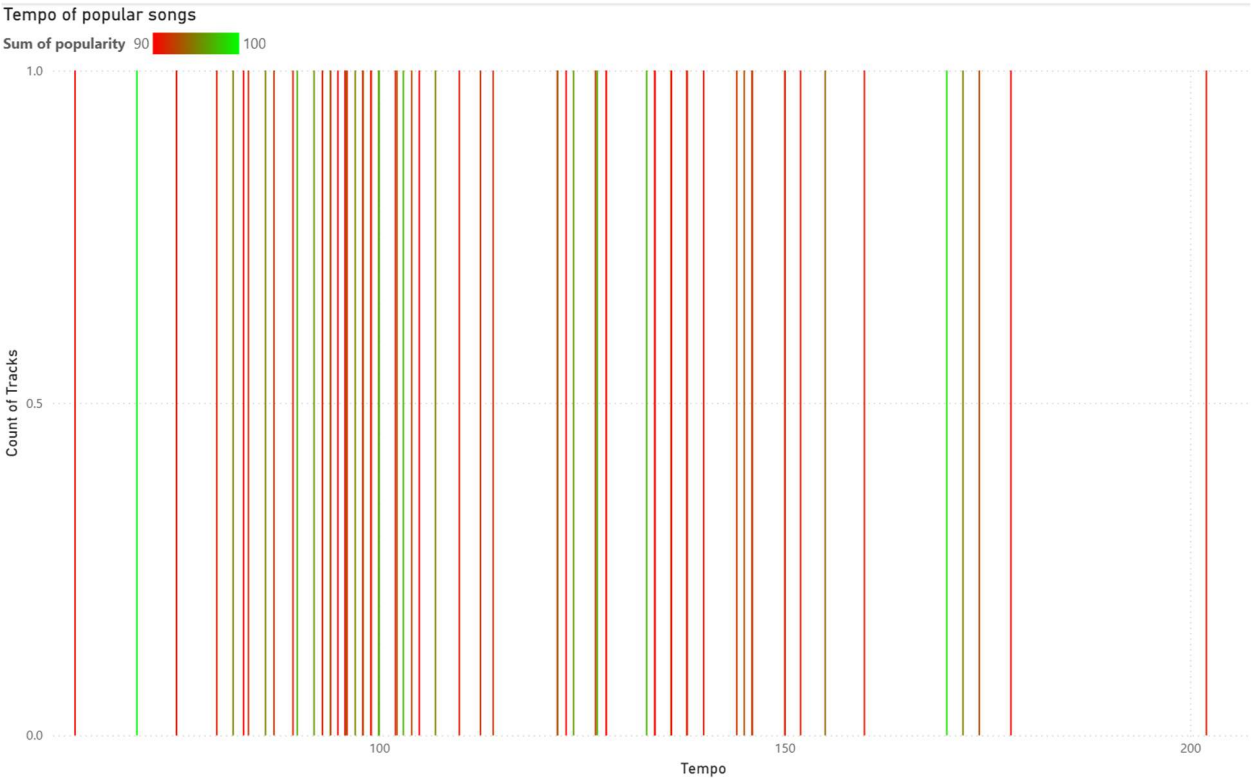
This visualization shows a generally positive relationship between danceability and popularity, with highly popular songs tending to be more danceable. While the relationship is not perfectly linear, the overall pattern supports what was observed during EDA and aligns with the model's feature importance results.

Popularity by Genre



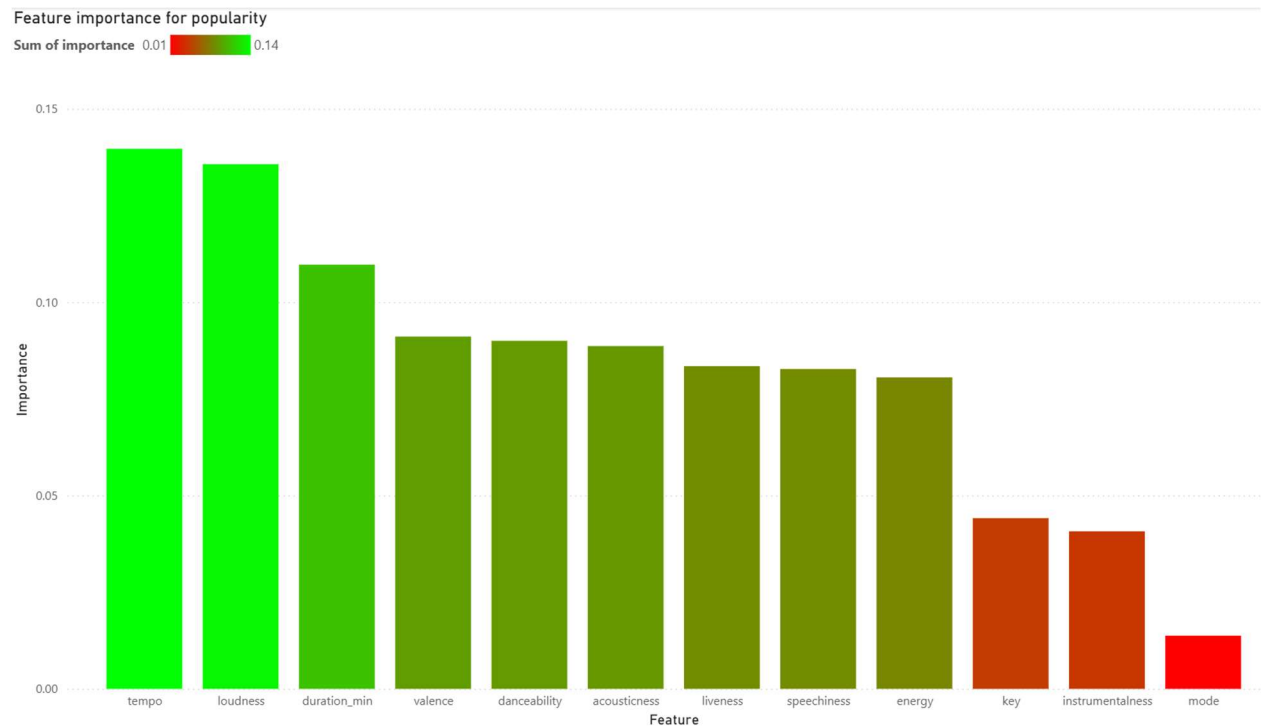
This chart highlights how popularity varies significantly across genres. Some genres consistently dominate in terms of aggregate popularity, while others remain more niche. This reinforced the idea that popularity is contextual and suggested that genre-specific modeling could be a useful next step.

Tempo Distribution of Popular Songs



Popular songs tend to cluster around a moderate tempo range, with fewer hits at extreme tempos. This visualization helped validate the model’s finding that tempo is one of the more influential predictors of popularity.

Feature Importance from the Model



This chart directly visualizes the RandomForest feature importance output. Tempo, loudness, duration, and danceability ranked highly, which closely matched insights found during exploration. Seeing this alignment across EDA, modeling, and visualization increased confidence in both the data pipeline and the analytical approach.

6. Recommendations and Next Steps

Based on the results of this project and the overall development experience, the next recommended steps would focus on deepening analysis rather than immediately expanding scope.

In the short term, adding model evaluation metrics and exporting predictions into Power BI would allow for more detailed performance analysis. Medium-term improvements could include dataset versioning, genre-aware models, and improved storage formats. In the longer term, the project could evolve into a more automated and scalable system by incorporating real-time ingestion and deploying the model as a service.

7. Conclusion

This project successfully demonstrates an end-to-end data engineering workflow for analyzing Spotify song popularity. More importantly, it provided a structured opportunity to explore the data, test assumptions, and validate findings across multiple layers of the pipeline. While the model itself is intentionally simple, the insights gained from the process highlight the value of combining data engineering, exploratory analysis, and machine learning in a single, cohesive system.